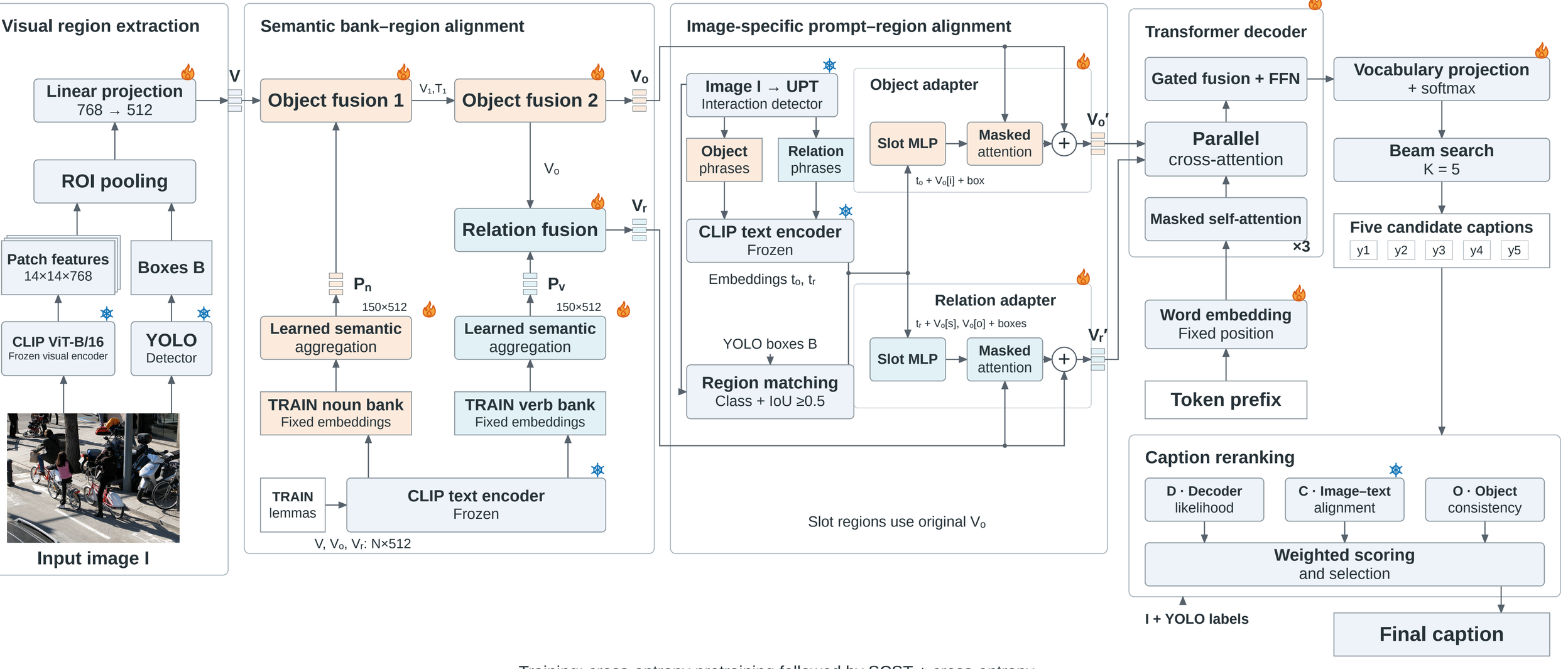

Visual region extraction
Linear projection
768 → 512
ROI pooling
Patch features
14×14×768
Boxes B
CLIP ViT-B/16
Frozen visual encoder
YOLO
Detector
Input image I
V
Semantic bank–region alignment
Object fusion 1
V1,T1
Object fusion 2
Vo
Vo
Relation fusion
Vr
Pn
150×512
Pv
150×512
Learned semantic
aggregation
Learned semantic
aggregation
TRAIN noun bank
Fixed embeddings
TRAIN verb bank
Fixed embeddings
TRAIN
lemmas
CLIP text encoder
Frozen
V, Vo, Vr: N×512
Image-specific prompt–region alignment
Image I → UPT
Interaction detector
Object
phrases
Relation
phrases
CLIP text encoder
Frozen
Embeddings to, tr
YOLO boxes B
Region matching
Class + IoU ≥0.5
Object adapter
Slot MLP
Masked
attention
to + Vo[i] + box
Vo'
Relation adapter
tr + Vo[s], Vo[o] + boxes
Slot MLP
Masked
attention
Vr'
Slot regions use original Vo
Caption generation
Transformer decoder
Gated fusion + FFN
Parallel
cross-attention
Masked self-attention
×3
Word embedding
Fixed position
Token prefix
Vocabulary projection
+ softmax
Beam search
K = 5
Five candidate captions
y1
y2
y3
y4
y5
Caption reranking
D · Decoder
likelihood
C · Image–text
alignment
O · Object
consistency
Weighted scoring
and selection
I + YOLO labels
Final caption
Training: cross-entropy pretraining followed by SCST + cross-entropy.
Frozen parameters
Trainable parameters